

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

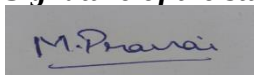
Constraint-Based Problem Solving

Code of Conduct:

I M.Prnai (192525068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWER

1. Hospital Doctor–Shift Assignment

Problem Understanding

There are 3 doctors: D1, D2, D3 and 3 shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

Constraint	Meaning
C1	D1 cannot work Night
C2	D2 must work before D3
C3	Only one doctor per shift
C4	D3 cannot work Morning
C5	Every doctor gets exactly one shift

Backtracking Process

Step	Assignment	Constraint Check	Result
1	D1 → Morning	D1 is not Night	✓ Valid
2	D2 → Afternoon	D2 is before D3	✓ Valid
3	D3 → Night	D3 is not Morning; D2 before D3	✓ Valid

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Constraint Verification

- D1 → Morning → ✓ D1 does not work Night
- D2 → Afternoon → ✓
- D3 → Night → ✓ D3 does not work Morning

- Only one doctor per shift $\rightarrow \checkmark$
- All doctors assigned exactly once $\rightarrow \checkmark$
- D2 works before D3 $\rightarrow \checkmark$

Final Answer

D1 $\rightarrow$ Morning, D2 $\rightarrow$ Afternoon, D3 $\rightarrow$ Night

Therefore, the assignment is a valid solution.

2. 5×5 Robot Grid – Search Process

Problem Understanding

A robot must move from Start (S) to Goal (G) in a 5×5 grid.

The assessment specifies:

- Movement: Up, Down, Left, Right
- Cost of each move = 1
- Obstacles cannot be crossed
- Manhattan Distance heuristic is required.

Manhattan Distance

The heuristic is:

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

where:

- (x_n, y_n) = current node
- (x_G, y_G) = goal position

Search Process

For every node:

1. Start from S.
2. Generate neighbouring cells.
3. Remove cells containing obstacles.
4. Calculate movement cost.
5. Calculate Manhattan distance.
6. Add/update nodes in the search structure.
7. Select the next suitable node.
8. Continue until G is reached.
9. Calculate total path cost.

Example Node Table

Node	g(n)	h(n)	Evaluation
S	0	Manhattan distance	Start
N1	1	Distance to G	1 + h
N2	2	Distance to G	2 + h
N3	3	Distance to G	3 + h
G	Final	0	Goal

3. Autonomous Rescue Robot

A. Constraint Analysis

Constraint	Effect on Decision Making
Up, Down, Left, Right movement	Robot has four possible movement actions
Obstacles	Robot cannot enter blocked cells
Limited sensing	Robot only knows adjacent cells
Normal movement cost = 1	Robot prefers shorter routes
Risky zone additional cost = 2	Robot should avoid unnecessary risky areas
Minimize total cost	Robot compares alternative paths
Dynamic knowledge	Robot updates its map while moving

Cost Calculation

Normal cell:

Cost = 1

Risky cell:

Cost = 1 + 2 = 3

Therefore, the robot should consider both **distance and risk**.

B. Solution Approach

Suitable Strategy

Uniform Cost Search (UCS) + Online Search Agent

Step-by-Step Process

Step	Process
1	Start at S
2	Observe adjacent cells

Step	Process
3	Identify blocked/risky/free cells
4	Generate possible movements
5	Calculate cumulative cost
6	Select the lowest-cost safe option
7	Move to the selected cell
8	Update knowledge about the environment
9	Repeat the search
10	Continue until survivor G is reached

Cost Example

Suppose the robot chooses:

$S \rightarrow A \rightarrow B \rightarrow G$

and all are normal cells:

$$1 + 1 + 1 = 3$$

If one cell is risky:

$$1 + 3 + 1 = 5$$

Therefore, the robot prefers the route with the **lowest total safe cost**.

C. Application of AI Concepts

1. Intelligent Agent

The robot is an intelligent agent because it:

Perceives $\rightarrow$ Decides $\rightarrow$ Acts $\rightarrow$ Learns/Updates

Example:

Observe obstacle $\rightarrow$ Select another path $\rightarrow$ Move $\rightarrow$ Update map

2. Rationality

A rational robot chooses the action that gives the best expected result.

It should:

- Avoid obstacles.
- Avoid unnecessary risky zones.
- Minimize movement cost.
- Reach the survivor safely.

- Update its knowledge when new information is discovered.

3. Environment Type

Property	Type
Observability	Partially observable
Agents	Single agent
Environment	Dynamic/uncertain
State	Grid-based
Movement	Discrete
Cost	Variable
Knowledge	Initially incomplete

D. Why UCS + Online Search is Suitable

UCS

UCS is suitable because:

1. Every movement has a cost.
2. Risky zones have additional cost.
3. The robot needs to minimize total cost, not simply number of moves.
4. UCS selects the lowest cumulative-cost path.

Online Search

Online search is suitable because:

1. The robot cannot see the complete environment.
2. It observes only adjacent cells.
3. New obstacles may be discovered during movement.
4. The robot can update its knowledge dynamically.